

Dictionaries and records

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Return values and local state

Without opening the old lesson, explain one key idea from Return values and local state. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lists and indexing

Without opening the old lesson, explain one key idea from Lists and indexing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace an update for Dictionaries and records. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Dictionaries and records. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Read, update and pass a dictionary record

Predict the three outputs and trace the named-field update. Explain why zero and a missing count produce different results. If the condition becomes if not count, which supplied record is first handled incorrectly?

```
def describe(record):  
    count = record.get("count")  
  
    if count is None:  
        return "missing"  
  
    return count  
  
record = {"count": 0}  
print(describe(record))  
  
record["count"] = 3  
print(describe(record))  
  
print(describe({}))
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Identify fields

For {"name": "Red", "capacity": 9, "used": 2}, state the keys, value of used and remaining spaces.

My reasoning and evidence

6. W2 Trace an update

Change the Red record used to 9. State remaining spaces and whether capacity changes.

My reasoning and evidence

7. W3 Repair a lookup

The code reads record["Used"] from the Red record. Explain the error and repair it without adding a field.

My reasoning and evidence

8. W4 Build a record independently

Create a Green record with capacity 15 and used 6. Pass its fields to remaining and store the result. Update used to 10 and repeat. Predict both results.

My reasoning and evidence

9. W5 Check the contract

A record has capacity 3 and used 4. It gives -1 spaces without an error. Should it be accepted? Explain.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled dictionaries and records evidence sheet to Python readiness portfolio. Show how you can read and update named dictionary fields. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Dictionaries and records

A zero field is still present. A dictionary record describes a shelf. The program distinguishes missing data from an actual zero.

Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

Dictionaries and records

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read a stored field

count exists with numeric value 0.



2. Read an absent field

get("price") returns None because price is missing and no default was supplied.



3. Test key membership

"count" in shelf is True even though its value is zero.

CHANGE ONE THING

Add the field price with value 0.

Explain the changed result using the matching lesson.

Predict all three original outputs. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A zero field is still present. A dictionary record describes a shelf. The program distinguishes missing data from an actual zero.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? "count" in shelf is True even though its value is zero.

1. Test key membership
2. Read an absent field
3. Read a stored field

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Test key membership "count" in shelf is True even though its value is zero. Dictionary keys name fields. Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

Practice 2: explain the deciding evidence

Changed-case practice: Add the field price with value 0. Which conclusion is supported by the supplied facts?

1. The price lookup now returns 0 instead of None.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The price lookup now returns 0 instead of None. A recorded zero is a present value, whereas the earlier lookup had no field.

Repair a missing building block

Review Return values and local state (python-bridge/9). Goal: Use a returned value in a later calculation. Short practice: Trace the two returned values and explain why first stays seven. What happens if the caller tries to print result directly?. Return to this week and explain what you changed.

Review Lists and indexing (python-bridge/7). Goal: Distinguish a list index from the value stored at that position. Short practice: Label the three index positions. Predict all original outputs, then name the first output and stopping point after using index 3.. Return to this week and explain what you changed.

Review Boolean conditions (python-bridge/4). Goal: Translate stated requirements into comparisons. Short practice: Give both comparison results for ages 11, 15 and 16. Which comparison rejects each outside value, and why does and matter?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
